MAS508-G Control Theory
Lecture 3: State Space Design

Prof. Jing Zhou

University of Agder

Department of Engineering Sciences

Room: D3 025 (UiA / Grimstad)

Phone: +47 3723 3191

Email: jing.zhou@uia.no

Website: http://www.uia.no/kk/prol/jingz

L3. State Space Design

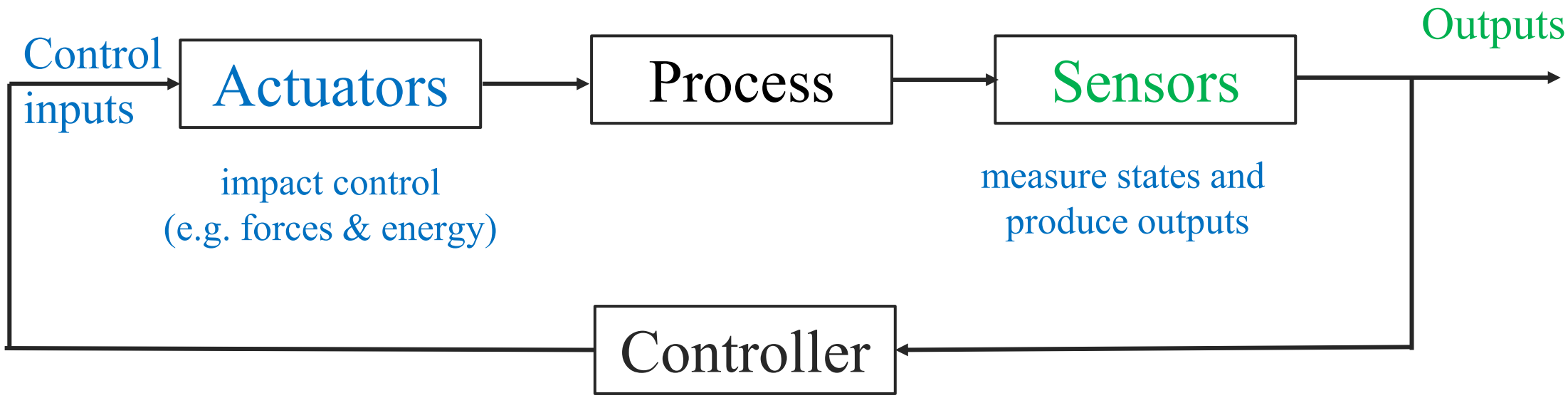
a) Time Response
b) Controllability
c) Observability
d) Pole Placement Control
e) Observer Design
f) Output Feedback Control
g) Optimal Control - Linear Quadratic Control (LQR)
h) Optimal Control – LQG

Textbook, “Control Systems Engineering”, Chapters 4 &12

L3.b Controllability

Book “Control Systems Engineering”, Chapter 12.3

Control



Controller design = how to use the sensor data (outputs) to generate the correct actuator commands (control inputs) to ensure that the outputs of the system satisfy the specifications.

What is controllability?

- If a system is controllable, then you can design a controller for it.



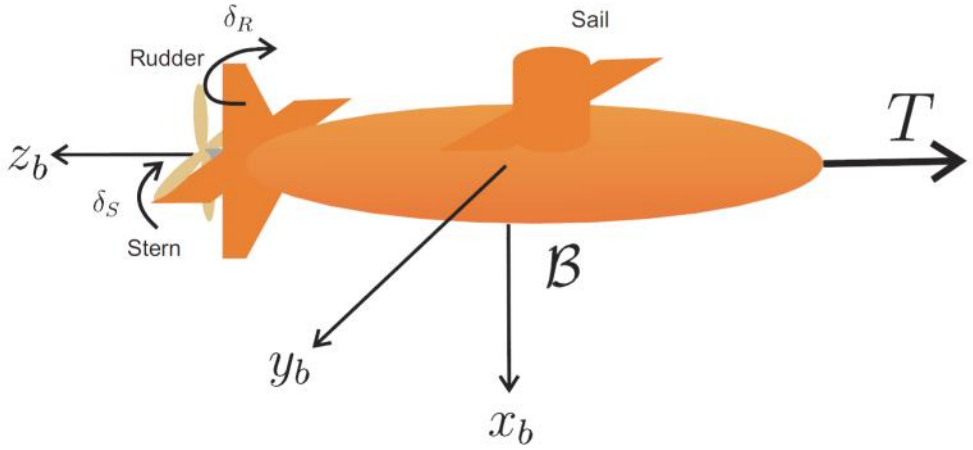
Pasteurization systems (Thermal process systems designed to optimize pasteurization of food, beverage, dairy, products) need to be controllable as they involve controlling the multiple elements:

- Temperature
- Level of liquid in a tank
- Flow rate

What determines controllability?

If a system does not have enough actuators or dose not have actuators in the correct configuration, then the system may not be controllable.

Example: If the Autonomous Underwater Vehicle (AUV) below did *not have a rudder or stern*, then it *would not be controllable*. You could not change its direction to reach the desired target position.



Controllability by inspection

- System 1: controllable
- Can place the poles of the closed-loop system where we desire.

$$\dot{x} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u$$

- System 2: uncontrollable
- Cannot place the poles of the closed-loop system where we desire.

$$\dot{x} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u$$

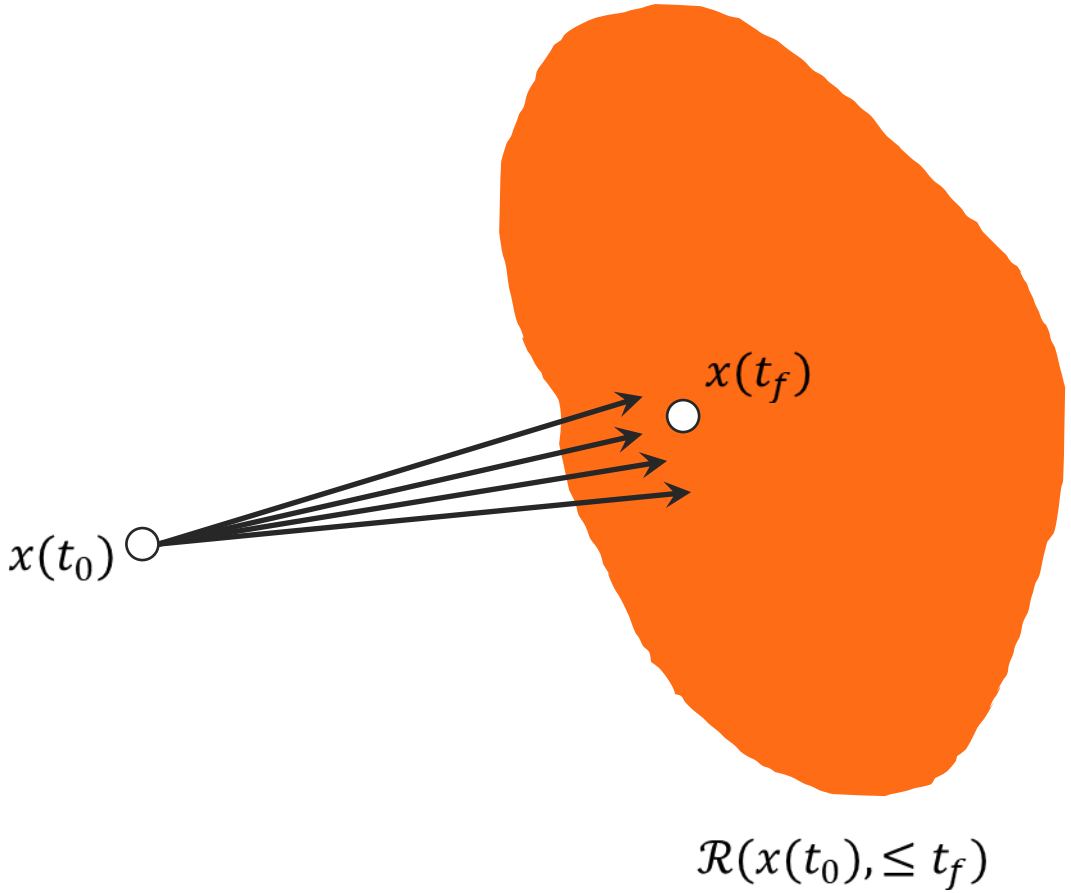
x_1 is not controlled by u

Controllability Definition

- If an input to a system can be found that takes every state variable from a desired initial state to a desired final state, the system is said to be **controllable**, otherwise, the system is uncontrollable.

- A system is ***controllable*** if the initial state $\boldsymbol{x(t_0)}$ can be driven to any other state $\boldsymbol{x(t)}$ in finite time using the control input $\boldsymbol{u(t)}$.

- When a system is *controllable*, then the states are said to be *reachable*.



Why do you need controllability?

- If a system is controllable, then you can design a controller that will make the system behave according to a certain desired behavior.

- If a system is controllable, you can design a state-feedback control, for example,

$$u = -Kx$$

for the system to achieve a certain desired behavior.

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Controllability Matrix

Recall the state-space model of a linear system:

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

where:

A = system matrix $(n \times n)$
B = input matrix $(n \times r)$
C = output matrix $(m \times n)$
D = feedforward matrix $(m \times r)$

The system state

$$x = [x_1 \quad x_2 \quad x_3 \ldots x_{n-1} \quad x_n]^T$$

and control input u.

The controllability matrix is defined as:

$$C_{ab} = [B \quad AB \quad A^2B \cdots A^{n-2}B \quad A^{n-1}B]$$

The system is completely state controllable **if and only if** the matrix is nonsingular.

Testing for Controllability

Rank Test:
The controllability matrix is nonsingular if and only if its rank equals the number of system states.

$$rank(C_{ab}) = n$$

where n = # of states in the system.

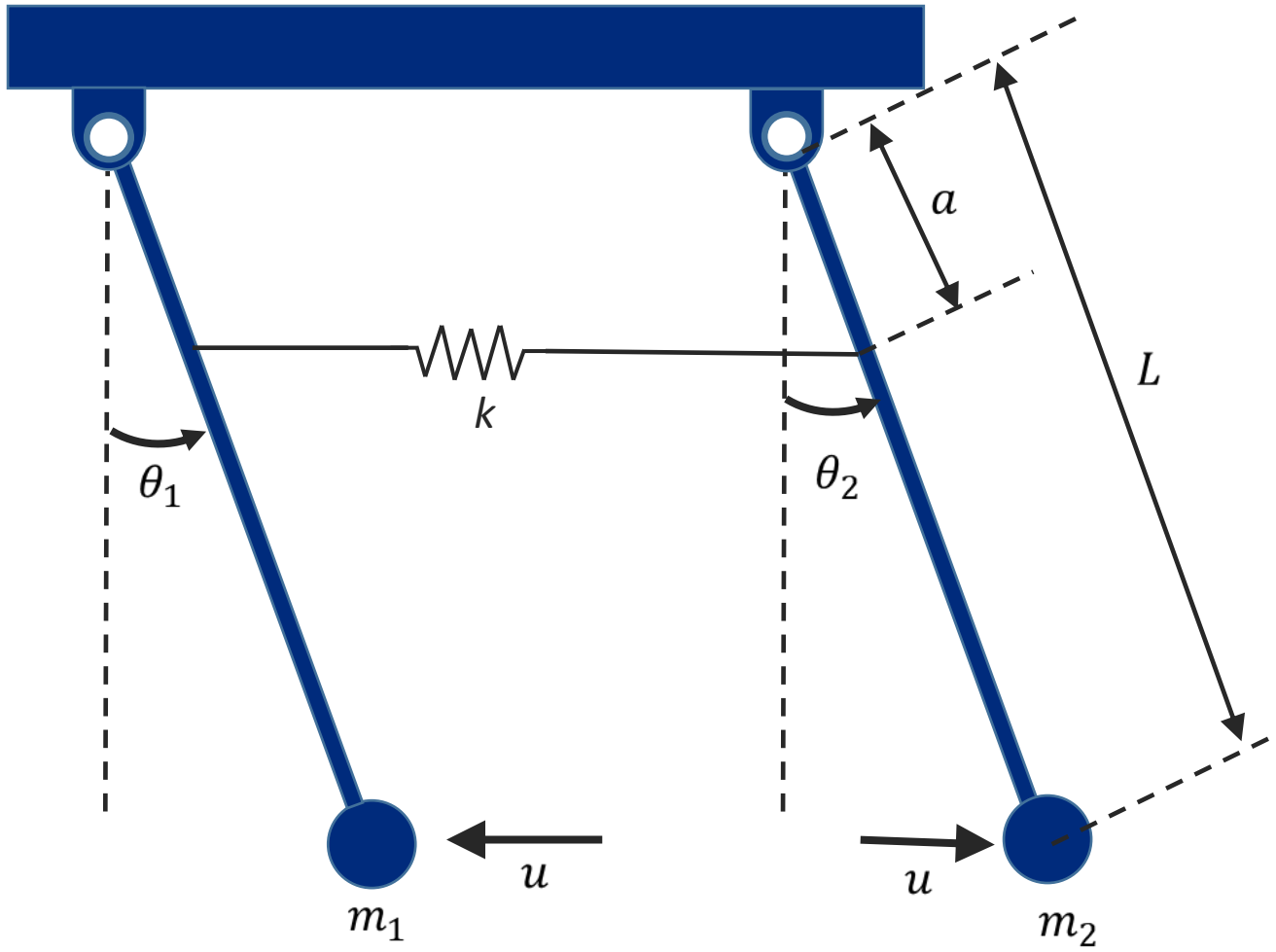
Determinant Test:
The controllability matrix is nonsingular if and only if its determinant is nonzero.

$$|C_{ab}| \neq 0$$

You can use either of these two methods to check if the system is full state controllable.

Determinant Test is only available for square matrix.

Example 1: Double Pendulum/Crane



Linear equations of motion:

$$mL^2\ddot{\theta}_1 = -ka^2(\theta_1 - \theta_2) - mgL\theta_1 - Lu$$
$$mL^2\ddot{\theta}_2 = -ka^2(\theta_2 - \theta_1) - mgL\theta_2 + Lu$$

State-space representation of system:

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{ka^2}{mL^2} - \frac{g}{L} & 0 & \frac{ka^2}{mL^2} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{ka^2}{mL^2} & 0 & -\frac{ka^2}{mL^2} - \frac{g}{L} & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ -1/(mL) \\ 0 \\ 1/(mL) \end{bmatrix} u$$

$$x = [\theta_1 \;\; \dot{\theta}_1 \;\; \theta_2 \;\; \dot{\theta}_2]$$

Is double crane controllable?

Given certain physical parameters:

$$\dot{x} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ -32.7 & 0 & 0.0063 & 0 \\ 0 & 0 & 0 & 1 \\ 0.0063 & 0 & -32.7 & 0 \end{bmatrix}}_{A} x + \underbrace{\begin{bmatrix} 0 \\ -27.8 \\ 0 \\ 27.8 \end{bmatrix}}_{B} u$$

Controllability matrix: $C_{ab} = [B \quad AB \quad A^2B \quad A^3B]$

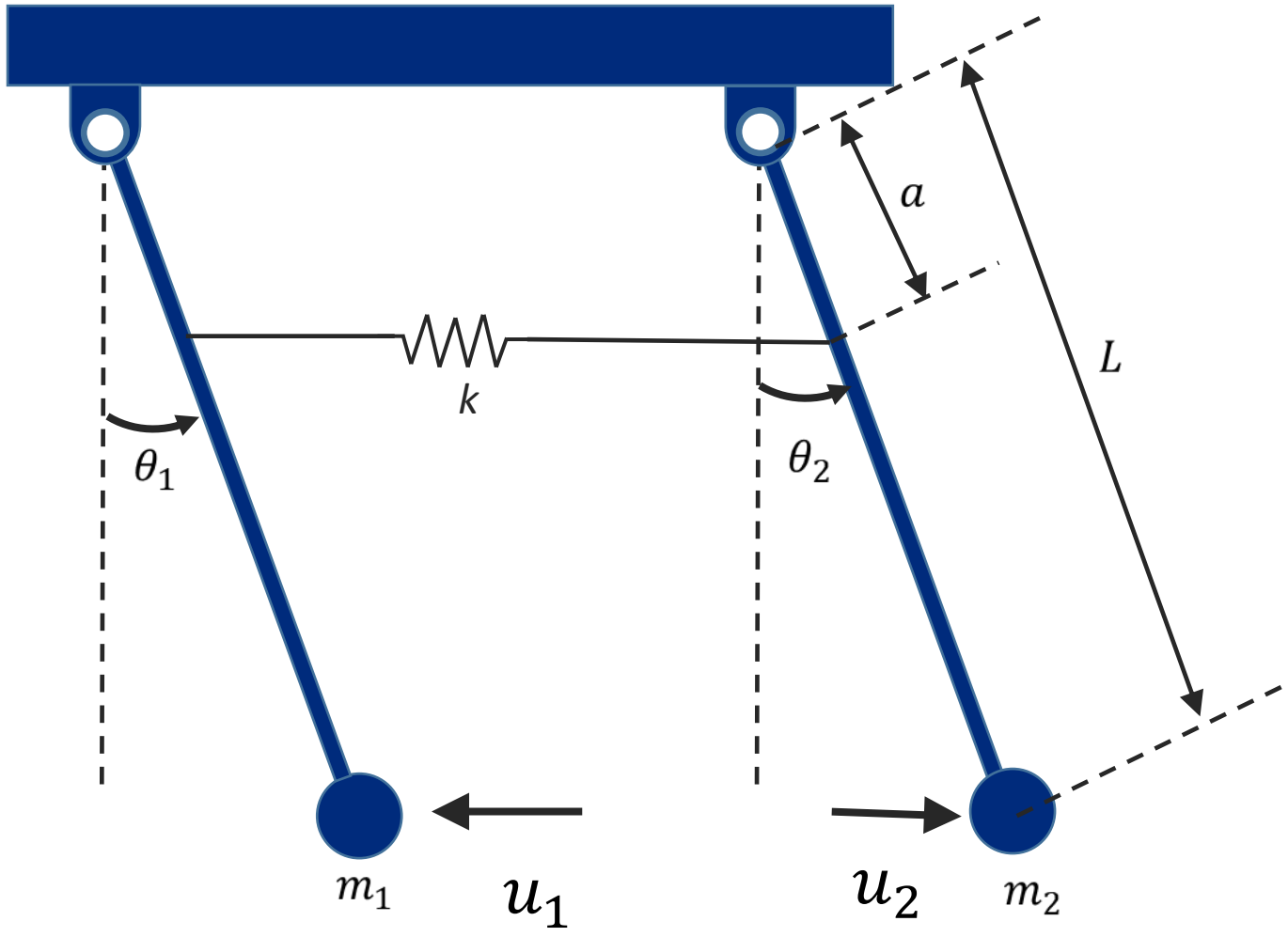
$$C_{ab} = \begin{bmatrix} 0 & -27.8 & 0 & 908.7 \\ -27.8 & 0 & 908.7 & 0 \\ 0 & 0 & 0 & -908.7 \\ 27.8 & 0 & -908.7 & 0 \end{bmatrix}$$

Determinant of controllability matrix:

$$|C_{ab}| = 0$$

So, the system is not controllable.

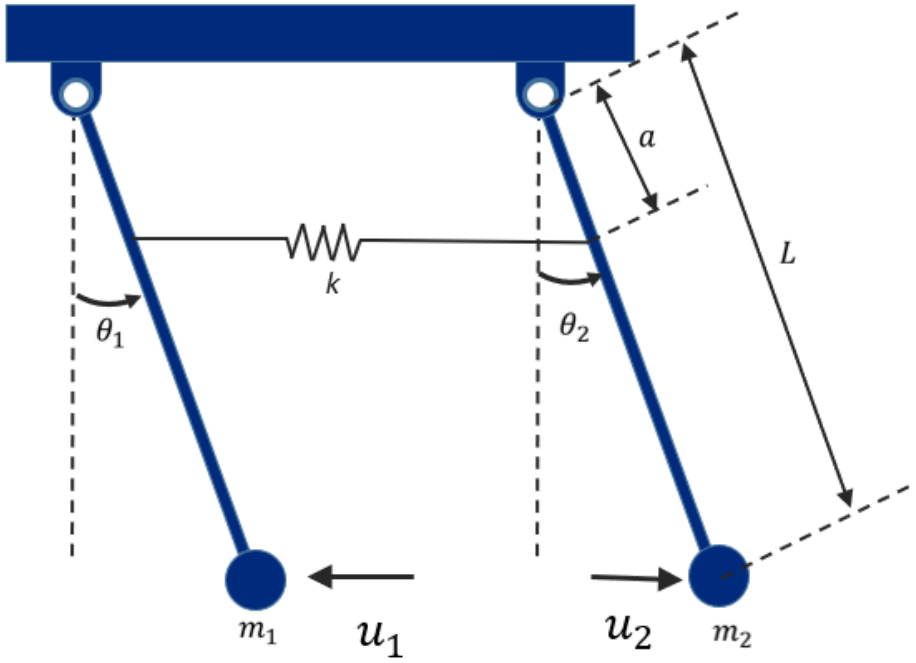
Is this what you expected?



Why is it not controllable?
Because there is only one input, *u*, to control each pendulum

How can we make it controllable?
Having different control input, u_1 and u_2, acting on each pendulum would solve this problem.

Example 2: Double Pendulum/Crane with two inputs



Linear equations of motion:

$$mL^2\ddot{\theta}_1 = -ka^2(\theta_1 - \theta_2) - mgL\theta_1 - Lu_1$$
$$mL^2\ddot{\theta}_2 = -ka^2(\theta_2 - \theta_1) - mgL\theta_2 + Lu_2$$

State-space representation of system:

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{ka^2}{mL^2} - \frac{g}{L} & 0 & \frac{ka^2}{mL^2} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{ka^2}{mL^2} & 0 & -\frac{ka^2}{mL^2} - \frac{g}{L} & 0 \end{bmatrix} x + \begin{bmatrix} 0 & 0 \\ -1/(mL) & 0 \\ 0 & 0 \\ 0 & 1/(mL) \end{bmatrix} u$$

Is double crane with two inputs controllable?

Given certain physical parameters:

$$\dot{x} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ -32.7 & 0 & 0.0063 & 0 \\ 0 & 0 & 0 & 1 \\ 0.0063 & 0 & -32.7 & 0 \end{bmatrix}}_{A} x + \underbrace{\begin{bmatrix} 0 & 0 \\ -27.8 & 0 \\ 0 & 0 \\ 0 & 27.8 \end{bmatrix}}_{B} u$$

Controllability matrix:

$$C_{ab} = [B \quad AB \quad A^2B \quad A^3B]$$

Rank of controllability matrix:

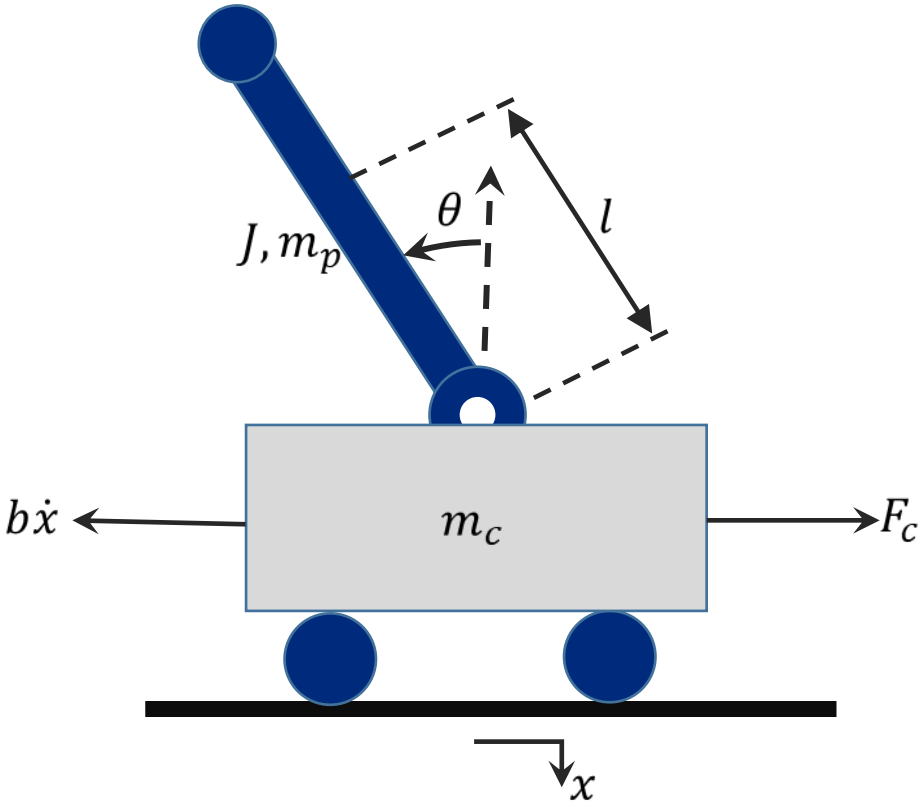
$$rank(C_{ab}) = 4$$

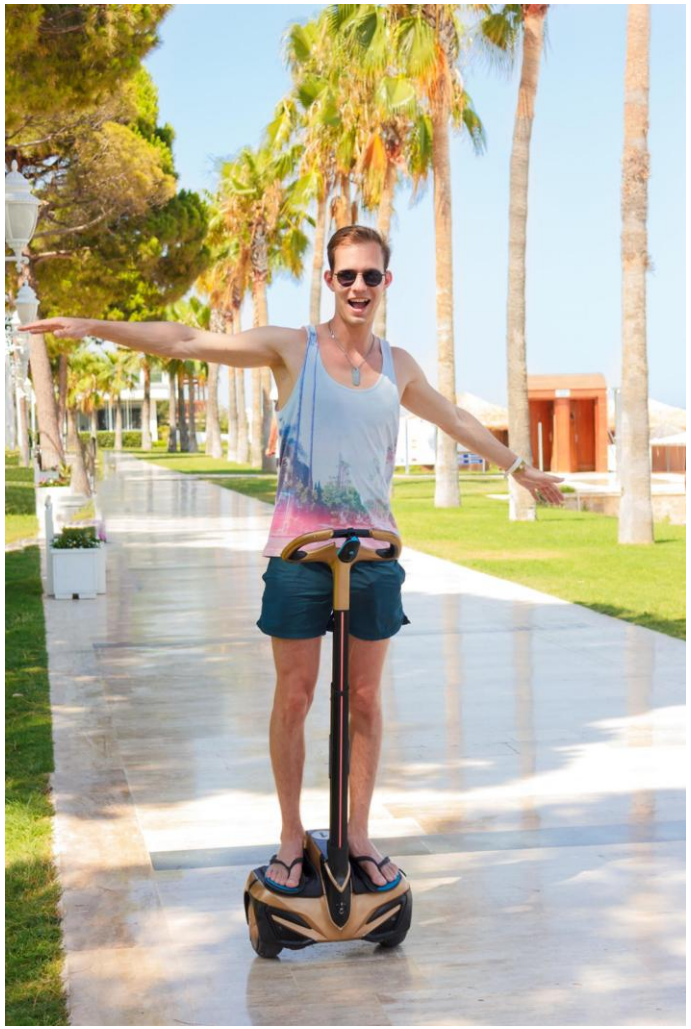
System is controllable.

```
% Controllability matrix
Cab = ctrb(A,B);
Rank_c=rank(Cab)
```

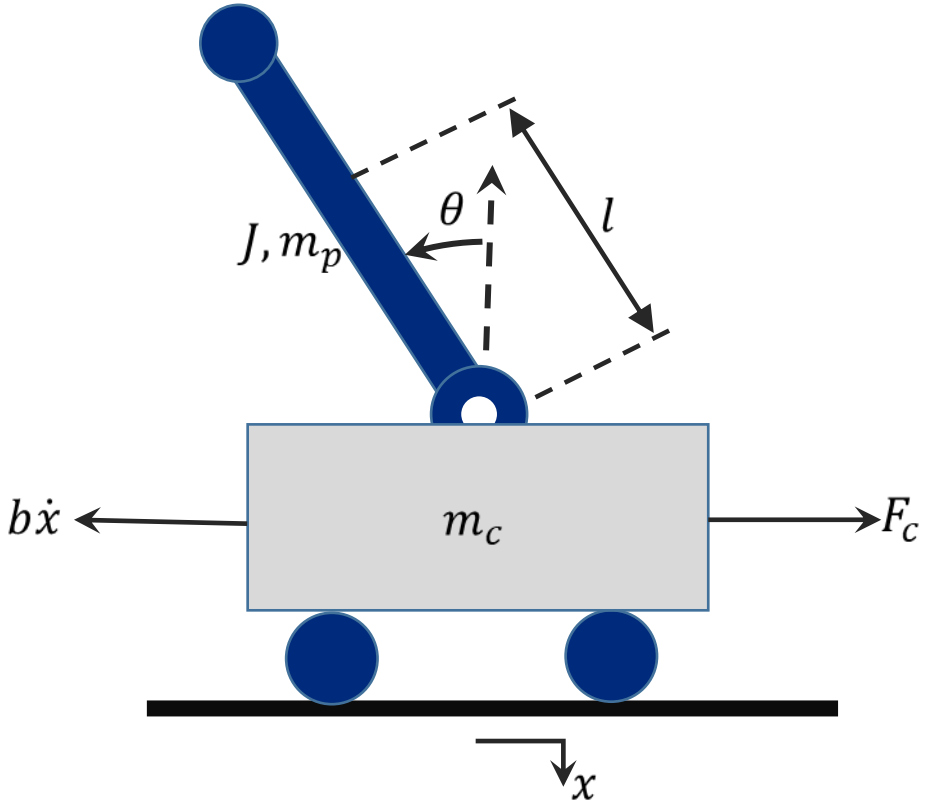
Example 3: Segway / Inverted Pendulum

The Segway system is similar to an inverted pendulum.





Model



Linear equations of motion:

$$(m_c + m_p)\ddot{x} = F_c - b\,\dot{x} + m_p l\,\ddot{\theta}$$

$$(J + m_p l^2)\ddot{\theta} = m_p l\ddot{x} + m_p g l\theta$$

Define the state and control input:

$$x^T = [x \quad \dot{x} \quad \theta \quad \dot{\theta}] \qquad u = F_c$$

Linear state-space representation (after solving EOMs):

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -(J + m_p l^2)b/J_t & m_p^2 g l^2/J_t & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -m_p l b/J_t & m_p g l(m_c + m_p)/J_t & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ (J + m_p l^2)/J_t \\ 0 \\ (m_p l)/J_t \end{bmatrix} u$$

Where

$$J_t = (J + m_p l^2)(m_c + m_p) - m_p^2 l^2$$

Is this linear inverted pendulum controllable?

Given certain physical parameters:

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -6.85 & 2.70 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -16.1 & 29.3 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 50.7 \\ 0 \\ 27.8 \end{bmatrix} u$$

Controllability matrix:

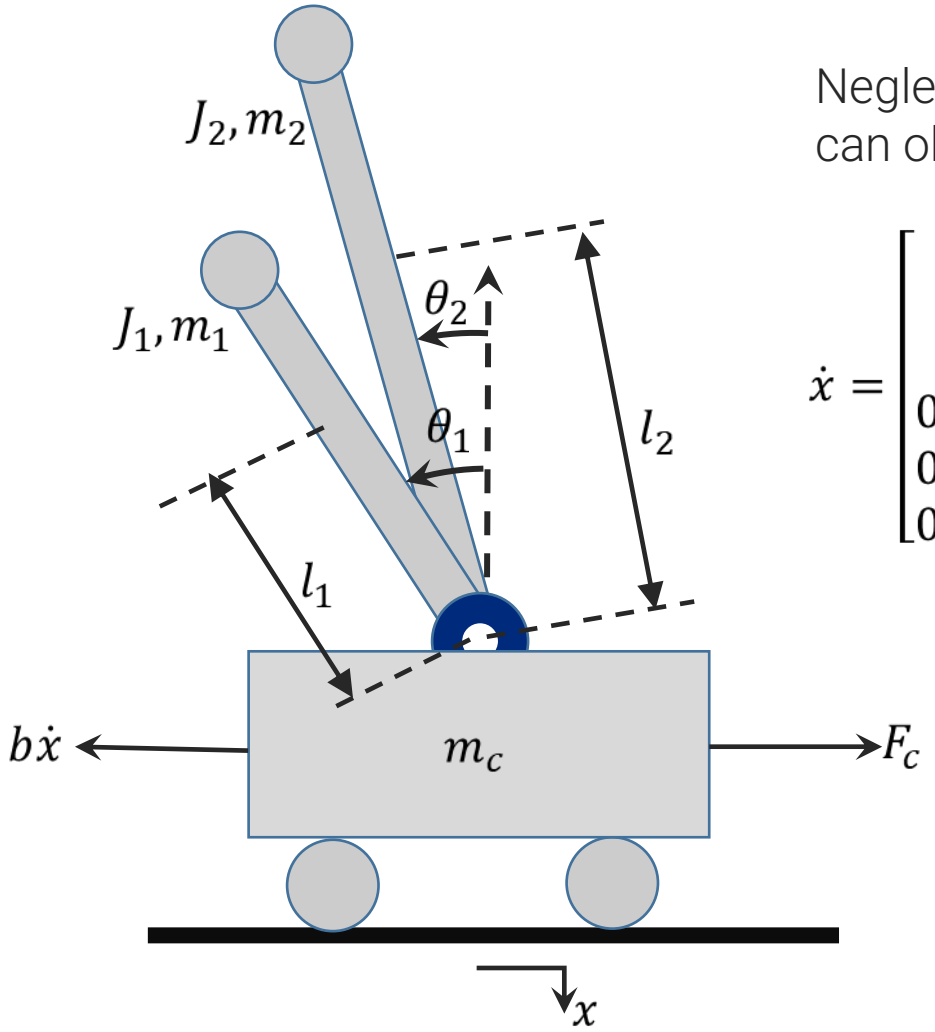
$$C_{ab} = [B \quad AB \quad A^2B \quad A^3B]$$

Rank of controllability matrix:

$$rank(C_{ab}) = 4$$

System is controllable.

Exercise: Dual/Parallel Inverted Pendulum



Can you balance both pendulums with one cart?

Neglecting the moment of inertia of the pendulum and any damping, we can obtain the following *simplified* state-space model:

$$\dot{x} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & m_1 g/m_c & m_2 g/m_c & 0 & 0 & 0 \\ 0 & (m_1+m_c)g/(l_1 m_c) & m_2 g/(l_1 m_c) & 0 & 0 & 0 \\ 0 & m_1 g/m_c l_2 & (m_1+m_c)g/(l_2 m_c) & 0 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/m_c \\ 1/(l_1 m_c) \\ 1/(l_2 m_c) \end{bmatrix} u$$

Is this controllable?

Case 1: Pendulums have different length and mass, i.e. $l_1 \neq l_2$

$$\dot{x} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 3.96 & 2.19 & 0 & 0 & 0 \\ 0 & 43.0 & 6.83 & 0 & 0 & 0 \\ 0 & 23.3 & 81.0 & 0 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1.75 \\ 5.48 \\ 10.32 \end{bmatrix} u$$

Controllability matrix:

$$C_{ab} = [B \quad AB \quad A^2B \quad A^3B \quad A^4B \quad A^5B]$$

Rank of controllability matrix:

$$rank(C_{ab}) = 6$$

System is **controllable.**

Case 2: Pendulums have same length and mass, i.e. $l_1 = l_2$

$$\dot{x} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 3.96 & 3.96 & 0 & 0 & 0 \\ 0 & 43.0 & 12.4 & 0 & 0 & 0 \\ 0 & 12.4 & 43.0 & 0 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1.75 \\ 5.48 \\ 5.48 \end{bmatrix} u$$

Controllability matrix:

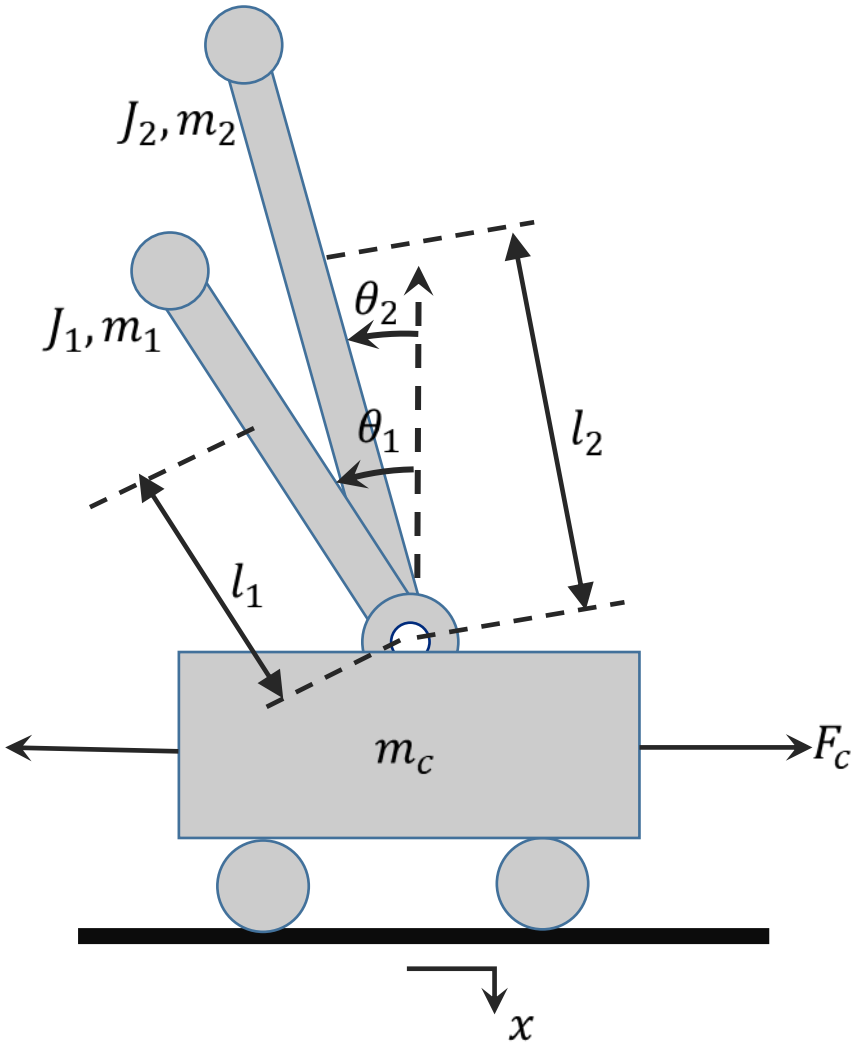
$$C_{ab} = [B \quad AB \quad A^2B \quad A^3B \quad A^4B \quad A^5B]$$

Rank of controllability matrix:

$$rank(C_{ab}) = 4 \neq 6$$

System is **not controllable.**

Is this what you expected?



Why is it not controllable when the pendulums are the same length?

Looking at the *B* matrix, having two identical pendulums basically means you have identical control input for both pendulums

How can we make it controllable?

Having pendulums with different length or a different mass.

Takeaways

Benefits

- If it's controllable, then you can design a controller for the system
- Can know if you need more or different type or different configuration of actuators

Drawbacks

- Computing controllability matrix can be tedious
- Need dynamic model of system that is accurate
- In many cases, you already know if a system can or cannot be controlled

Partial Controllable
- Controllable/Uncontrollable Decomposition

Stabilizability

- Weaker than controllability
- A system is *stabilizable*, when all uncontrollable state variables can be made to have stable dynamics.(Its unstable modes are controllable.)
- Thus, even though some of the state variables cannot be controlled (as determined by the controllability test above) all the state variables will still remain bounded during the system's behavior.

Example:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}1\\ 0\end{bmatrix}u$$

The unstable eigenvalue at $\lambda_1 = 1$ can be controlled with control input *u*.

The uncontrollable state x_2 at $\lambda_2 = -1$ is already stable.

Definitions Summary

- Controllability: System is controllable if all the states can be reached to desired locations using control inputs.
- Stabilizability: System is stabilizable when all uncontrollable state variables can be made to have stable dynamics.

Controllable/Uncontrollable Decomposition

- **Note that we** only establish the ***equivalence*** relationship between ***arbitrary placement of*** the closed-loop poles and the controllability of the system ($\boldsymbol{A}$, $\boldsymbol{B}$).

- In other words, if it is ***not*** required to ***arbitrarily place*** the closed-loop poles, the system ($\boldsymbol{A}$, $\boldsymbol{B}$) is not necessarily controllable.
- If the system is not controllable, the system can be decomposed into Controllable and Uncontrollable states.

Example: Uncontrollable Decomposition

Example. Consider the following system in state-space. Find the controllable/uncontrollable decomposition.

$$\dot{\boldsymbol{x}} = \left[\begin{array}{cc:cc} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ \hdashline 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{array}\right] \boldsymbol{x} + \left[\begin{array}{c} 0 \\ 1 \\ \hdashline 1 \\ 1 \end{array}\right] u$$

- **Solution.** Check the controllability matrix

$$C_{ab} = [B \quad AB \quad A^2B \quad A^3B] = \begin{bmatrix} 0 & 1 & 4 & 12 \\ 1 & 2 & 4 & 8 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

$\text{rank}(\boldsymbol{C_{ab}}) = 3 < 4$, the system is uncontrollable.

- Introduce the transformation matrix $\boldsymbol{P}$ for controllability decomposition.

$$P^{-1} = [B \quad AB \quad A^2B \quad q_1] = \begin{bmatrix} 0 & 1 & 4 & 0 \\ 1 & 2 & 4 & 0 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 0 \end{bmatrix}$$

where the first 3 columns are any 3 *independent* columns in $\boldsymbol{C_{ab}}$, and the remaining column $\boldsymbol{q_1}$ **is arbitrarily chosen so that $\boldsymbol{P^{-1}}$ is nonsingular.**

$$P^{-1} = \begin{bmatrix} 0 & 1 & 4 & 0 \\ 1 & 2 & 4 & 0 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 0 \end{bmatrix} \qquad P = \begin{bmatrix} -2/3 & 5/9 & 0 & 4/9 \\ -1/3 & 4/9 & 0 & -4/9 \\ 1/3 & -1/9 & 0 & 1/9 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

- Then the equivalence transformation $\bar{x} = Px$ transforms

$$\dot{\bar{x}} = P\,\dot{x} = P(Ax + Bu) = PAP^{-1}\,(Px) + PB\,u = \bar{A}\,\bar{x} + \bar{B}\,u$$

- Compute $\bar{A}, \bar{B}$

$$\bar{A} = PAP^{-1} = \left[\begin{array}{ccc:c} 0 & 0 & -4 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 0 \\ \hdashline 0 & 0 & 0 & -1 \end{array}\right] = \begin{bmatrix} \bar{A}_{11} & \bar{A}_{12} \\ 0 & \bar{A}_{22} \end{bmatrix} \quad \bar{B} = PB = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \bar{B}_{11} \\ 0 \end{bmatrix}$$

- Then the equivalence transformation $\bar{x} = Px$ transforms

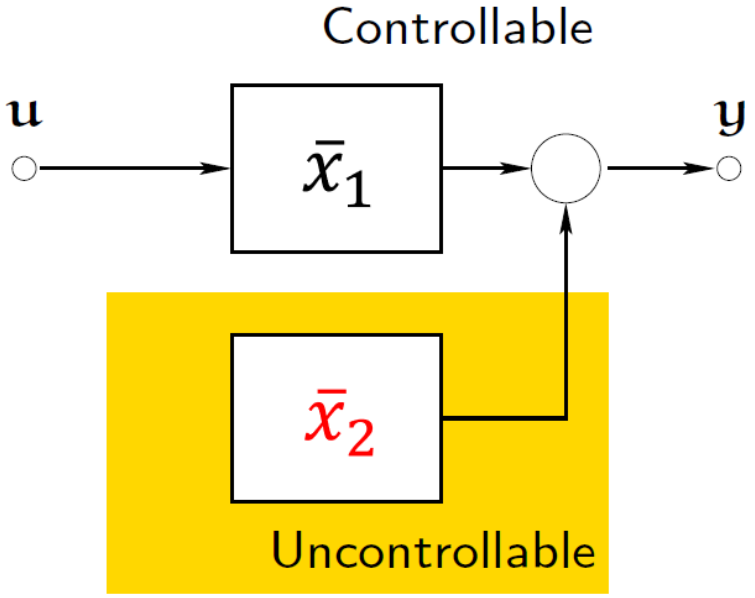
$$\dot{\bar{x}} = P\,\dot{x} = P(Ax + Bu) = PAP^{-1}\,(Px) + PB\,u = \bar{A}\,\bar{x} + \bar{B}\,u$$

$$\dot{\bar{x}} = \begin{bmatrix} \dot{\bar{x}}_1 \\ \dot{\bar{x}}_2 \end{bmatrix} = \bar{A}\,\bar{x} + \bar{B}\,u = \left[\begin{array}{c|c} \bar{A}_{11} & \bar{A}_{12} \\ \hline 0 & \bar{A}_{22} \end{array}\right] \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} + \begin{bmatrix} \bar{B}_1 \\ 0 \end{bmatrix} u$$

- The states in the new coordinates are decomposed into

$\bar{x}_1$: 3 *controllable states*

$\bar{x}_2$: 1 *uncontrollable state*

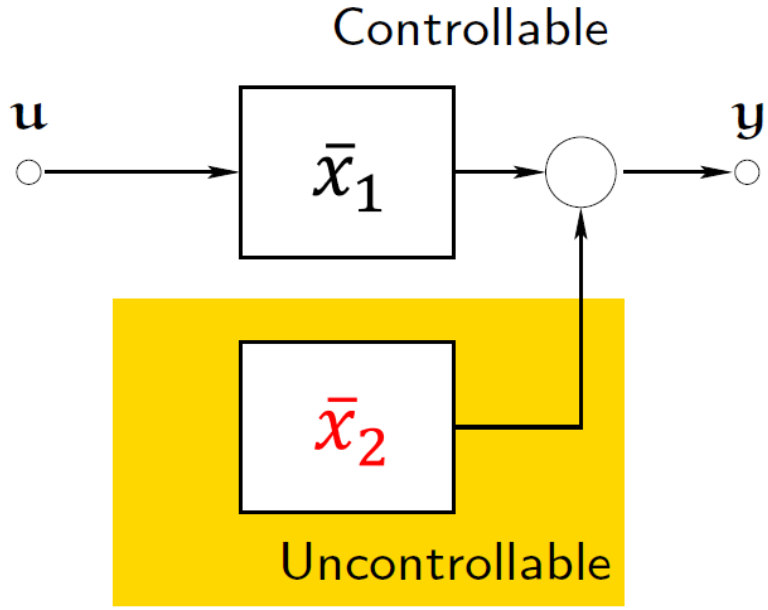


Note: The uncontrollable state $\bar{x}_2$ at $\bar{\lambda}_2 = -1$ is stable.

- The states in the new coordinates are decomposed into

$\bar{x}_1$: 3 *controllable states*

$\bar{x}_2$: 1 *uncontrollable state*



- The reduced-order state equation of the controllable states

$$\dot{\bar{x}}_1 = \bar{A}_{11}\bar{x}_1 + \bar{B}_1 u$$

is controllable.

The MATLAB function ctrbf transforms a state equation into its controllable/uncontrollable staircase form.

$$\bar{A} = PAP^{-1} = \begin{bmatrix} 0 & 0 & -4 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & \mathbf{-1} \end{bmatrix} = \begin{bmatrix} \bar{A}_{11} & \bar{A}_{12} \\ 0 & \bar{A}_{22} \end{bmatrix} \quad \bar{B} = PB = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \bar{B}_{11} \\ 0 \end{bmatrix}$$

- This indicates there is one eigenvalue (-1) of $\boldsymbol{A}$ cannot be altered no matter how the control gain $\boldsymbol{k}$ is chosen.

- In other words, there is always a closed-loop located at -1 no matter how $\boldsymbol{k}$ in feedback control $u = -\boldsymbol{kx}$ is chosen.

- **Case I:** It is ***impossible*** to find the gain k in $u = -\boldsymbol{kx}$ to place the closed-loop poles at $-2, -3, -4, -1.5$

- **Case II:** $-2, -3, -4, \mathbf{-1}$. It is ok.